



Airoha IoT SDK for BT Data HID

Release Notes

Version: 1.1.1
Release date: 30 Apr 2026

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Document Revision History

Revision	Date	Description
1.0.0	3 December 2025	Initial release.
1.0.1	15 January 2026	Feature improvement.
1.1.0	17 Apr 2026	Added new supported features for the game controller and PC tools.
1.1.1	30 Apr 2026	Added new features.

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1 Introduction

The Airoha IoT Software Development Kit (SDK) for BT Data provides a comprehensive software solution. The SDK supports hardware abstraction layers (HAL), peripheral drivers, Zephyr RTOS, and other features.

1.1 Architecture Layout of the SDK

Figure 1 and Figure 2 show the two-layer architecture layout of the SDK for AB162x, which includes Applications, Middleware, and Zephyr.

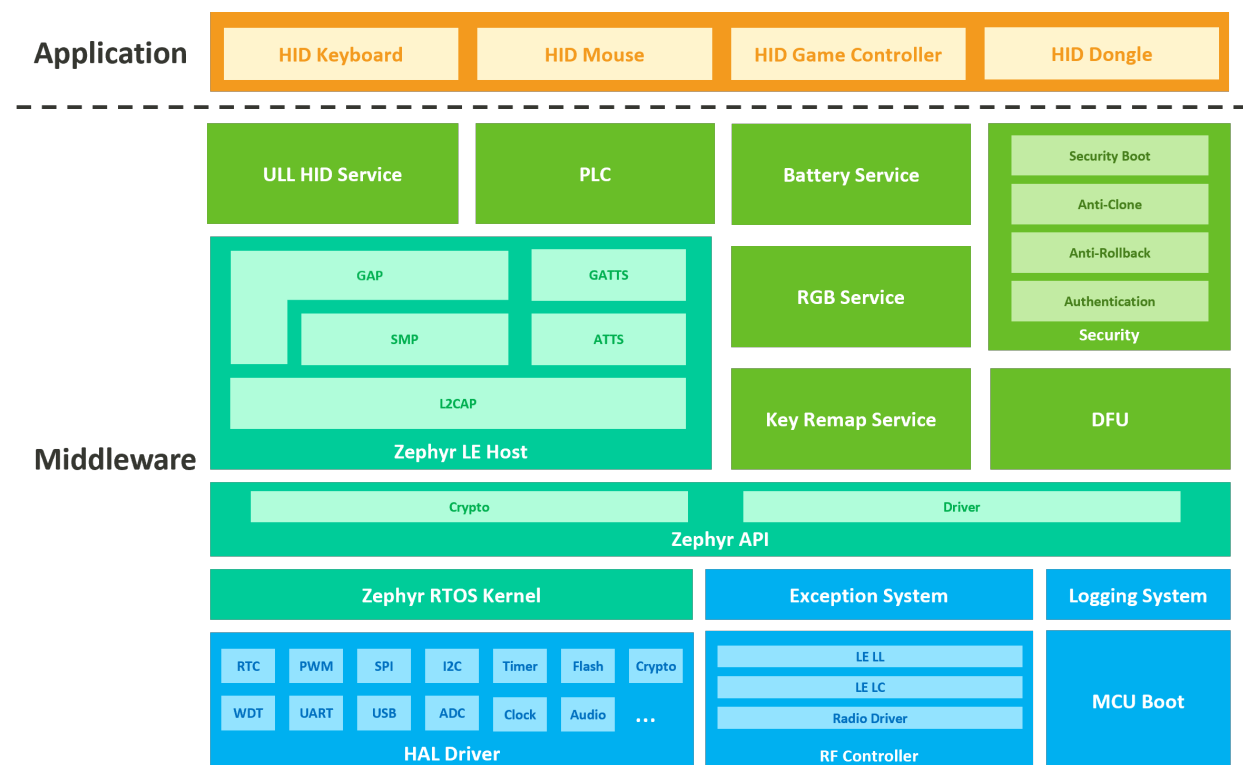


Figure 1. SDK Architecture for BT Data - HID

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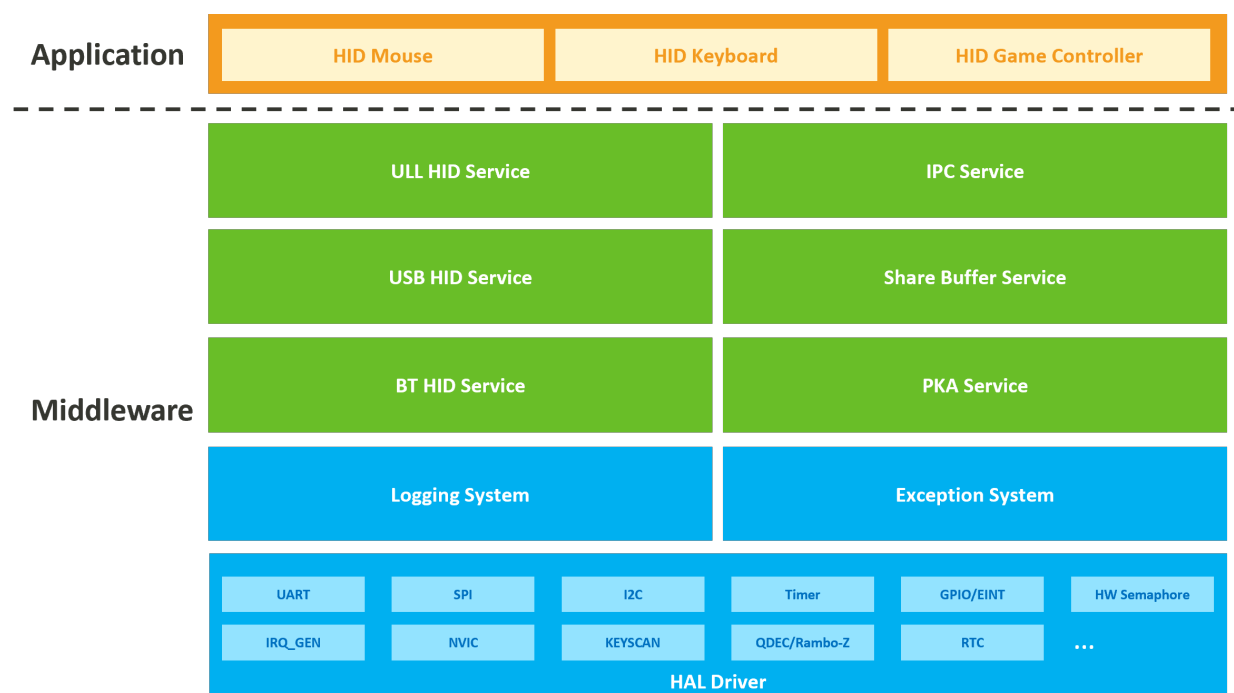


Figure 2. SDK Architecture for RISC-V

The top layer includes the application projects running on the SDK. They are based on the Middleware, OS, and HAL layers. These layers provide rich features for application development. For example, the Middleware layer provides the Bluetooth Low Energy Stack. The Zephyr layer provides the underlying real-time operating system.

1.2 Knowledgebase

The SDK includes documentation to show developers each module and provides an explanation of the features in a convenient and developer-oriented approach.

The following documentation is available with the SDK.

- *Airoha IoT SDK for BT Data Build Environment Guide* – This document provides information about how to create and build a project, and how to create a module with the SDK in the BT Data build environment.
- *Airoha IoT SDK for BT Data Firmware Update Developer's Guide* – This document shows how FOTA operates and how you can adjust the memory usage of FOTA.
- *Airoha IoT SDK for BT Data <project> User Guide* – This document aims to assist users in comprehending and effectively utilizing the functionality of the man-machine interface (MMI) layer in the example project.
- *Airoha IoT SDK for BT Data and Smart Home Get Started Guide* – This guide covers the SDK features, step-by-step setup of the development environment, and its usage.
- *Airoha IoT SDK for BT Data and Smart Home Open Source Software Guide* – This document explains the open source modules and the features used in the SDK.
- *Airoha IoT SDK for BT Data and Smart Home Memory Layout Developer's Guide* – This document provides details about the memory layout of the SDK, and how to adjust the memory layout for custom applications.

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- *Airoha IoT SDK for BT Data and Smart Home Power Mode Developer's Guide* – This document addresses the MCU system's power mode configuration and power consumption measurement focused on power modes provided by the Airoha IoT SDK.
- *3rd-party Software Component Guide* – This guide is a reference for the third-party components bundled in the SDK, including open-source software (OSS) and business software (BS). It provides descriptions of each module, license information, software versions, and links to official websites.
- *Airoha RACE Command Specification* – This document provides information on the format of RACE commands, including the type, bytes, and value, along with examples of their usage and important notes.

Table 1-1 shows the documents related to BT Data that are under the <sdk_root>/airoha/mcu/doc folder.

Table 1-1. Documentation Relevance for BT Data

Document File Name	Airoha IoT SDK for BT Data (HID)
Airoha_IoT_SDK_for_BT_Data_HID_Release_Notes.pdf	√
Airoha_IoT_SDK_for_BT_Data_Build_Environment_Guide.pdf	√
Airoha_IoT_SDK_for_BT_Data_Firmware_Update_Developers_Guide.pdf	√
Airoha_IoT_SDK_for_BT_Data_HID_Device_Reference_Design_User_Guide.pdf	√
Airoha_IoT_SDK_for_BT_Data_and_Smart_Home_Get_Started_Guide.pdf	√
Airoha_IoT_SDK_for_BT_Data_and_Smart_Home_Memory_Layout_Developers_Guide.pdf	√
Airoha_IoT_SDK_for_BT_Data_and_Smart_Home_Open_Source_Software_Guide.pdf	√
Airoha_IoT_SDK_for_BT_Data_and_Smart_Home_Power_Mode_Developers_Guide.pdf	√
3rd-party_Software_Component_Guide.html	√
Airoha_RACE_Command_Specification.html	√

1.3 Terms and Definitions

The terms and definitions used in this release note are listed in the following table.

Table 1-2. Terms and Definitions

Terms	Definitions
DFU	Device Firmware Update
FOTA	Firmware Over-The-Air

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2 SDK Version 1.1.1

2.1 Main Changes

- Software features:
 - Supports dynamic power control with VPA adjustment.
 - Fixed the bug related to dynamic 4M / 2M PHY switching.
 - Gaming mouse enhancements:
 - Updated HID mouse report format:
 - Sensor data resolution increased from 12 bits to 16 bits.
 - Report size expanded from 7 bytes to 9 bytes in 2.4G mode.

2.2 Known Issues

There is a known issue with this SDK version.

- Gaming Mouse
 - **Compatibility Issue during DFU / FOTA Upgrade from V1.0.1 to V1.1.1**
 - When performing upgrades via DFU / FOTA, there are certain compatibility limitations between the two versions:
 - In version V1.0.1, the mouse transmits X and Y data as 12 bits, with a report format of 7 bytes.
 - If the mouse is upgraded to V1.1.1 first while the dongle remains at V1.0.1, the dongle will not be able to recognize the new report data sent by the mouse, resulting in abnormal communication.
 - If the dongle is upgraded to V1.1.1 first, it can receive and parse both 7-byte (12 bits) and 9-byte (16 bits) report formats, thus supporting both V1.0.1 and V1.1.1 mouse devices.

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3 SDK Version 1.1.0

3.1 Main Changes

- Software features:
 - Gaming Controller
 - Added support for 2.4G mode (8K / 4K / 2K / 1K / 500 / 250 / 125 report rate).
 - Added support for USB mode (8K / 4K / 2K / 1K / 500 / 250 / 125 report rate).
 - Added support for BT mode (133 report rate).
 - Added support for Real-Time RGB and Flash RGB.
 - Added support for key remap and marco keys.
 - Added support for endpoint in / out with INT mode.
 - Added support for DFU and FOTA.
 - Added support for Xinput and Dinput.
 - Added support for Dynamic TX Power Adjustment and Dynamic 4M / 2M PHY Switching.
 - PC Tools
 - Added support for FOTA.

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4 SDK Version 1.0.1

4.1 Main Changes

- Software features:
 - Gaming Mouse
 - Reduced the reconnection time after standby.

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5 SDK Version 1.0.0

5.1 Main Changes

- Software features:
 - Gaming Mouse
 - Added support for 2.4G mode (8K / 4K / 2K / 1K / 500 / 250 / 125 report rate).
 - Added support for USB mode (8K / 4K / 2K / 1K / 500 / 250 / 125 report rate).
 - Added support for BT mode (133 report rate).
 - Added support for Real-Time RGB and Flash RGB.
 - Added support for key remap and marco keys.
 - Added support for endpoint in / out with INT mode.
 - Added support for Device Firmware Update (DFU).
 - Added support for Packet Loss Concealment for sensor data.
 - Added support for mechanical buttons and optical buttons.
 - Added support for mechanical wheel and optical wheel.
 - PC Tools
 - Added support for features such as report rate switch, macro key, profile setting, debounce time, sensor LOD, and motion sync setting.
 - Added support for DFU.
 - Added support for Real-time RGB control in UT App.

5.2 Known Issues

There is a known issue with this SDK version.

- Gaming Mouse
 - Due to controller changes below 8K report rate, the previous firmware version needs to switch to an 8K report rate before performing a DFU upgrade to this firmware version. Additionally, both the mouse and dongle must be updated to the latest version to enable use at report rates below 8K.

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